



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 11



By- Satish Sir

Recap of Previous Lecture



Topic

Connectivity in Graph Part-03



Topics to be Covered



Topic

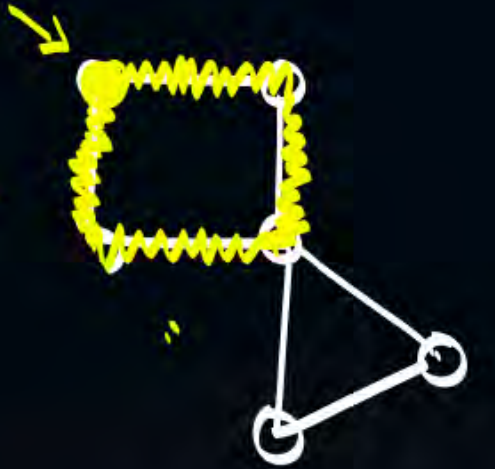
Connectivity in Graph Part-04

Topic

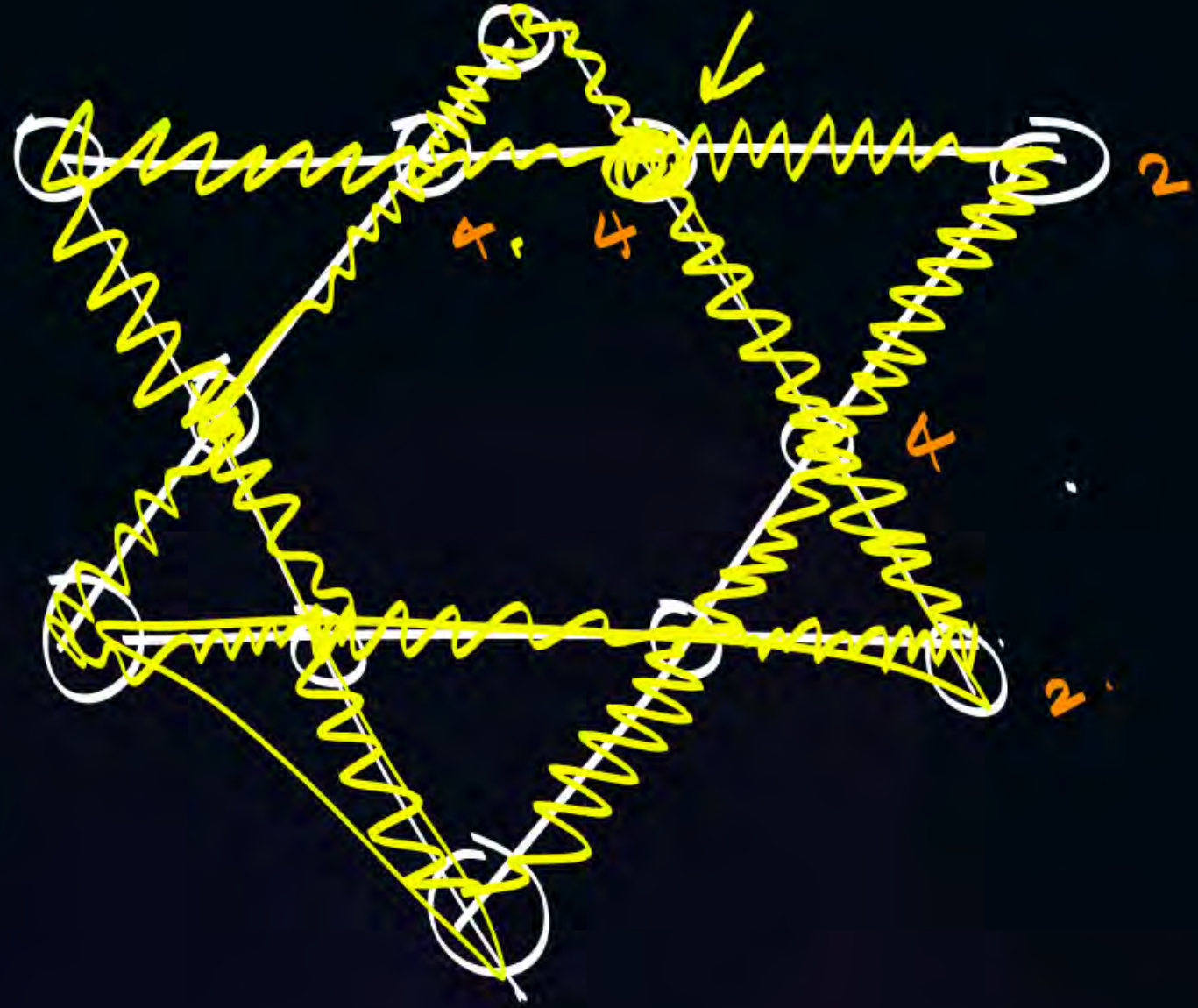
Chromatic Number

Trail:
alternating sequences of vertices & edges $R.v \mid R \setminus E$

Closed Trail: Trail + starting = ending vertices.

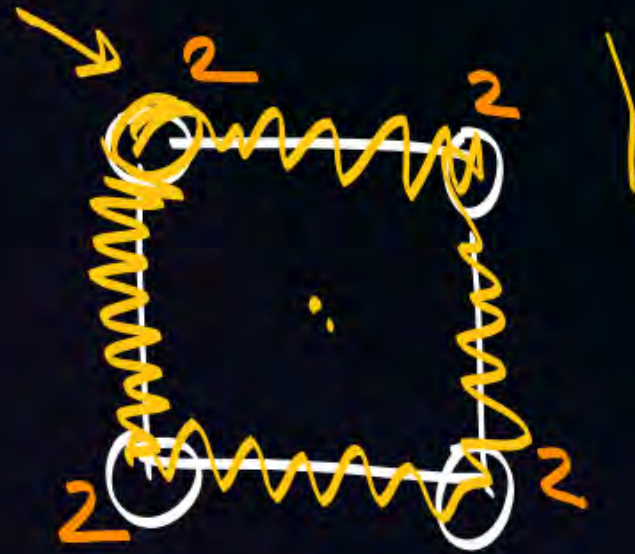


Euler ckt: Closed Trail + all the edges exactly once.





closed Trail.
+
cover all
the edges
exactly once.

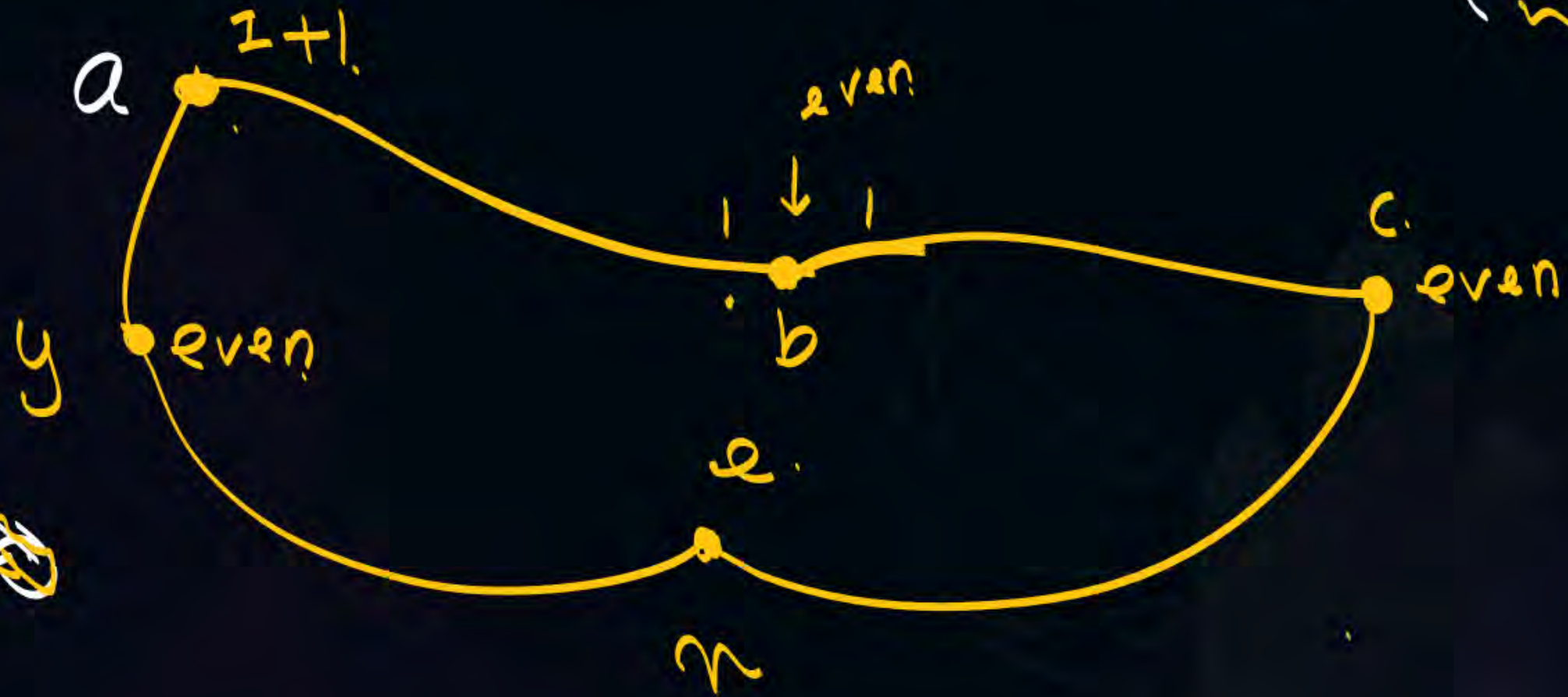


Graph.
↓
Euler ckt
↓
Euler Graph.



Thm: Graph is Euler Graph if and only if degrees of all vertices are even. (connected)

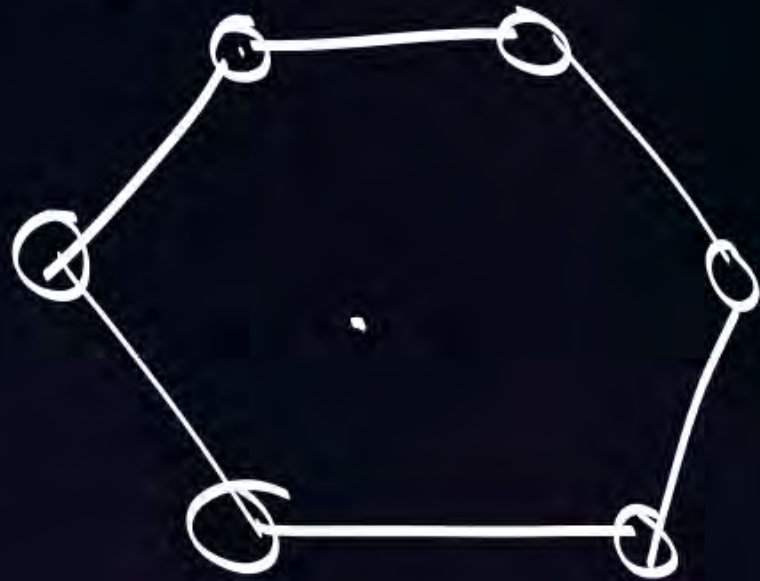
St = a. ending = a. all edges.



(false)

Euler Graph exist 6 vertices graph with all degrees are 2.

E.g.



C.G.



$n = 6$
 $K = 2$

Trail:

open Trail:

Euler Path. = { open Trail
OR
Euler line. +
covers all edges
exactly once.

1) point

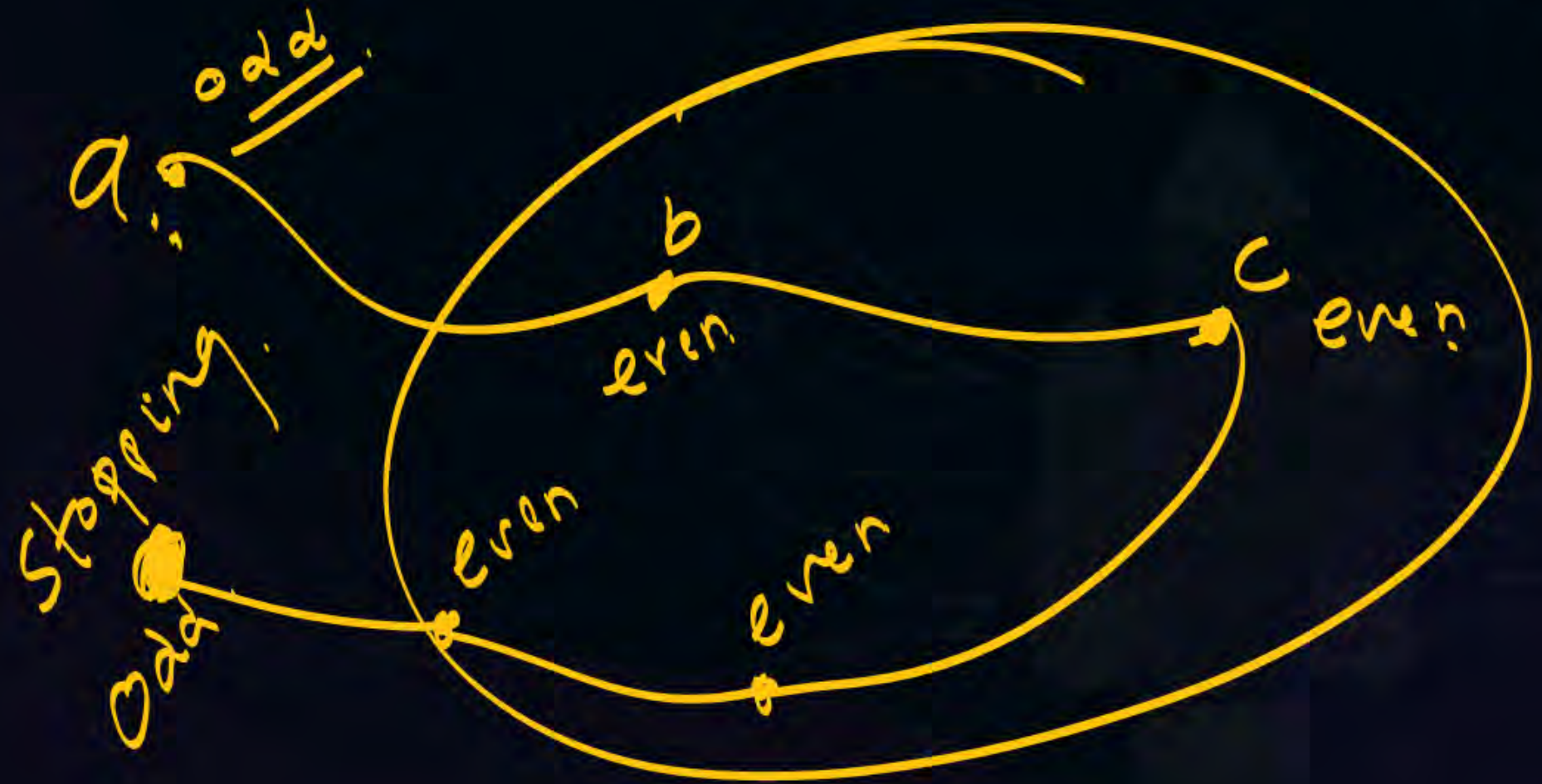
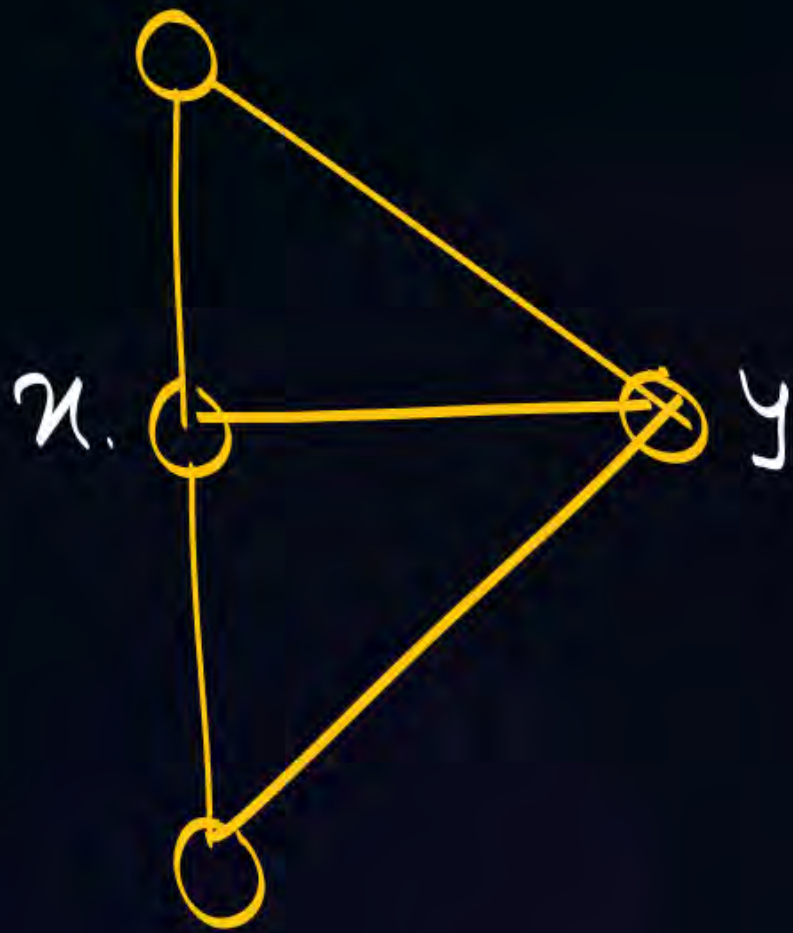
* 2) all edges.

$St = x$
 $end = y$

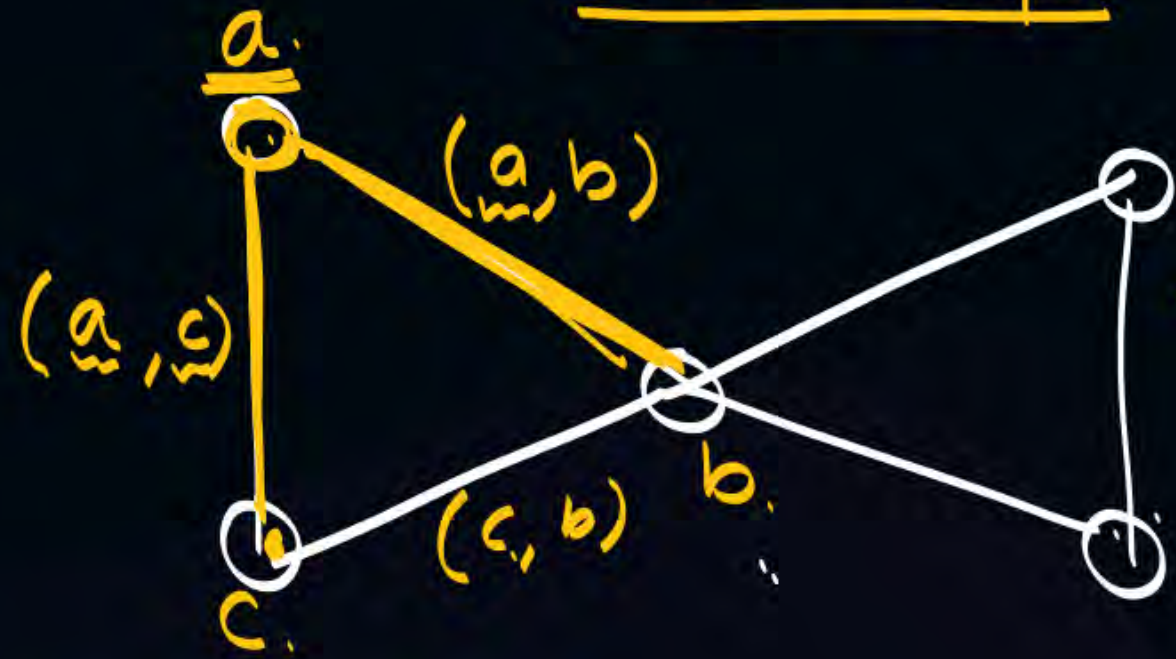


Graph contains Euler Path if and only if Graph.

Start: odd vertex. contains exactly
End: odd vertex. 2 odd degree
vertices.

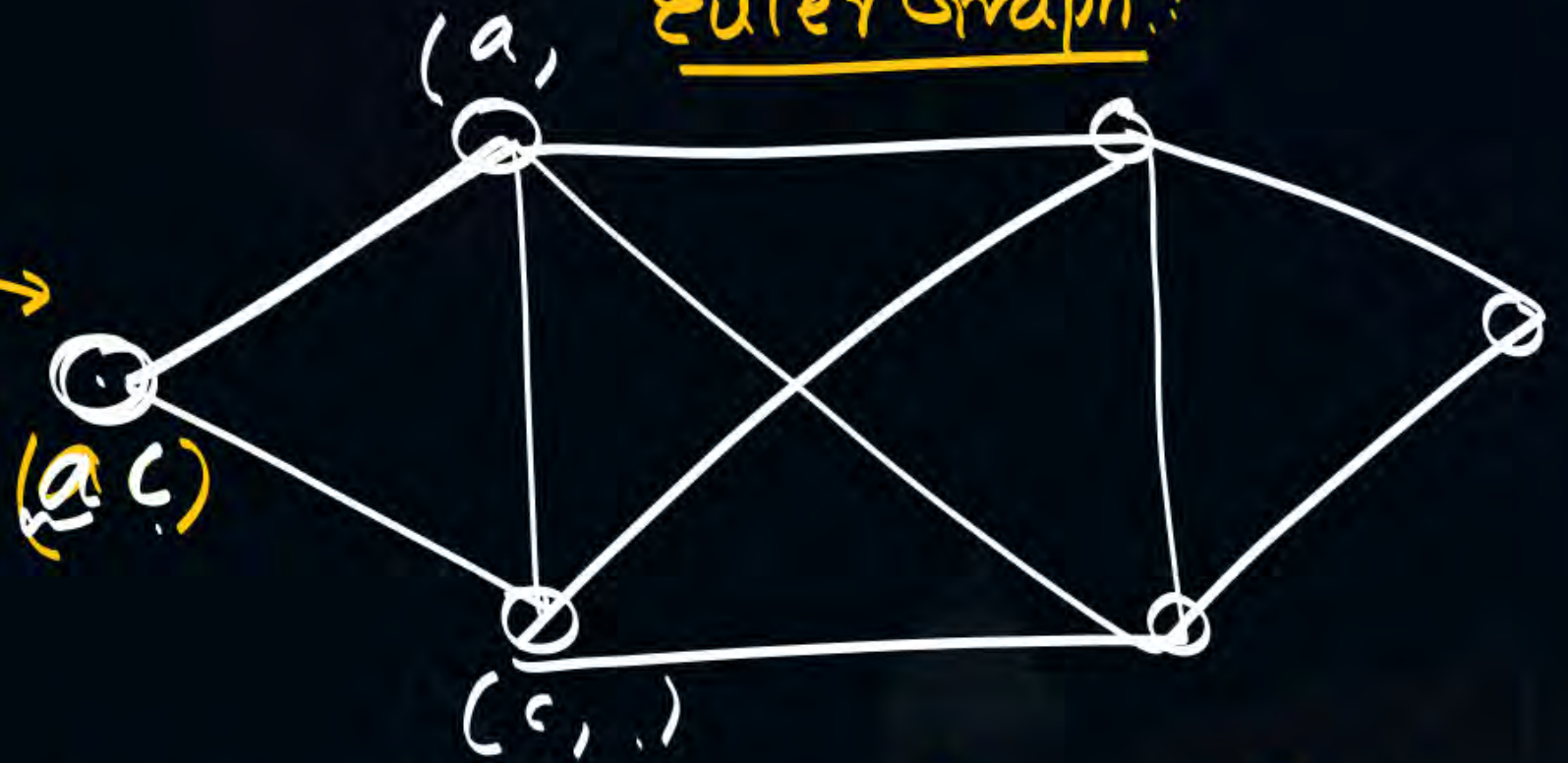


Euler Graph:



$L(G)$

Euler Graph:



Degree of a: 2 Line Graph of Every Euler Graph will also be Euler Graph.

covering.

all edges.

closed Trail.

open Trail.

Degrees of
all are
even

Euler ckt

Euler Graph.

Euler path

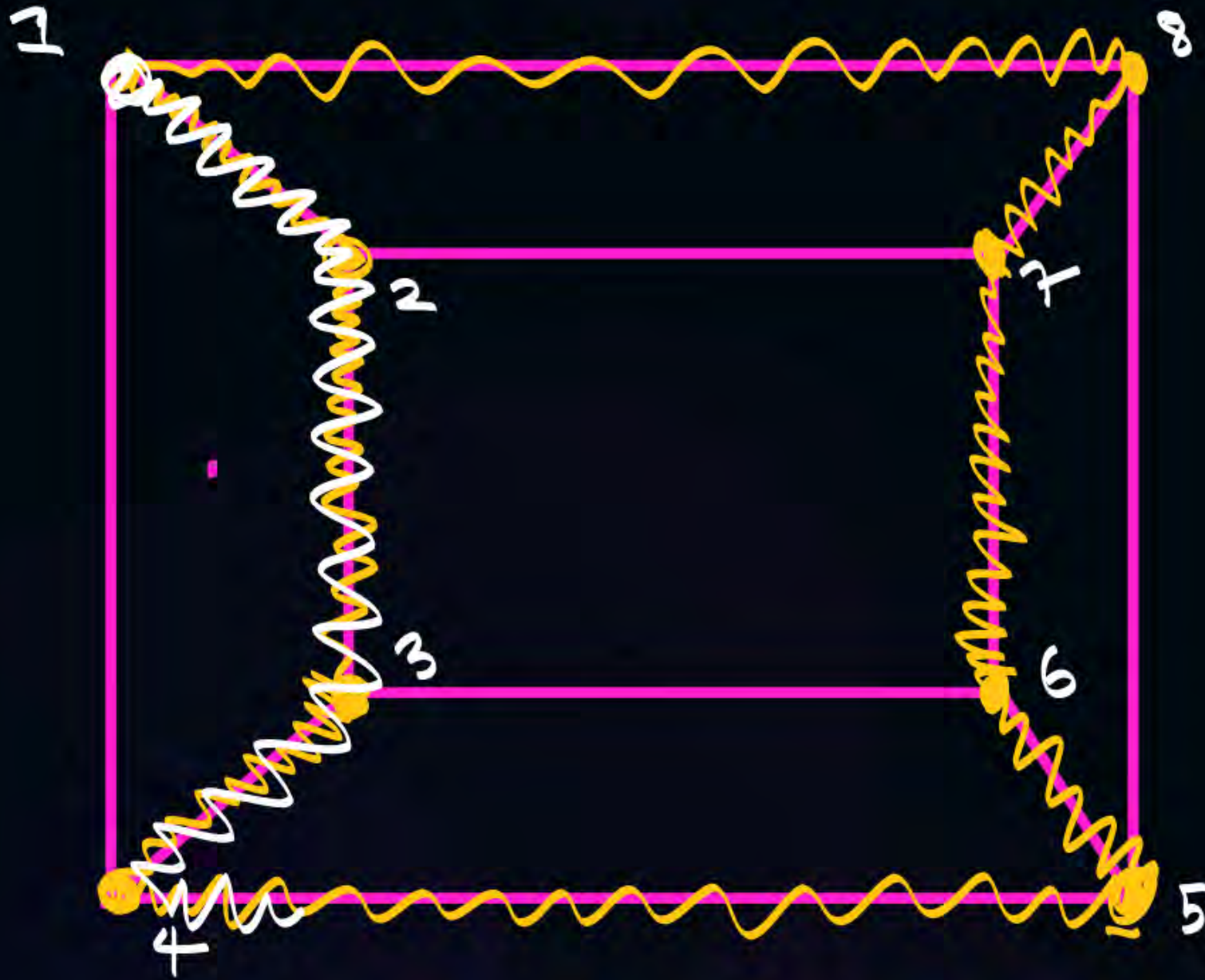
Exactly
2 odd degree
vertices.

Path :

Closed Path : Path + st = ending vertices.

Hamiltonian ckt : closed path.
+

cover all the vertices
exactly once.



Hamiltonian ckt

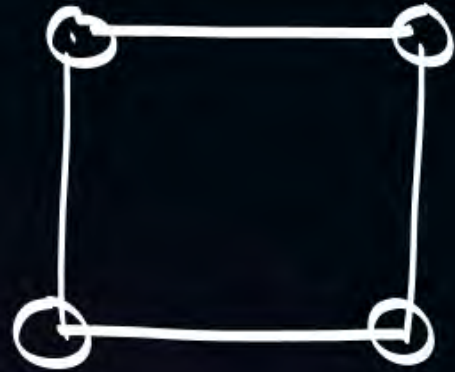
$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8$

Graph $\rightarrow$ Hamiltonian ckt



Hamiltonian Graph. $G = (V, E)$

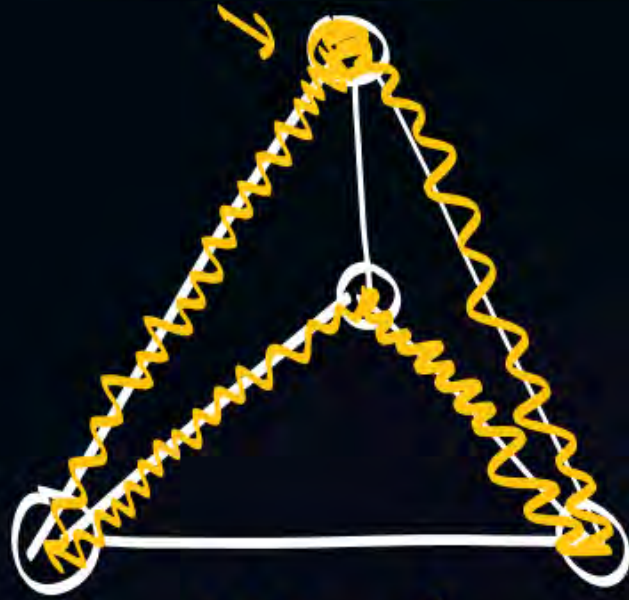
Cycle Graph:



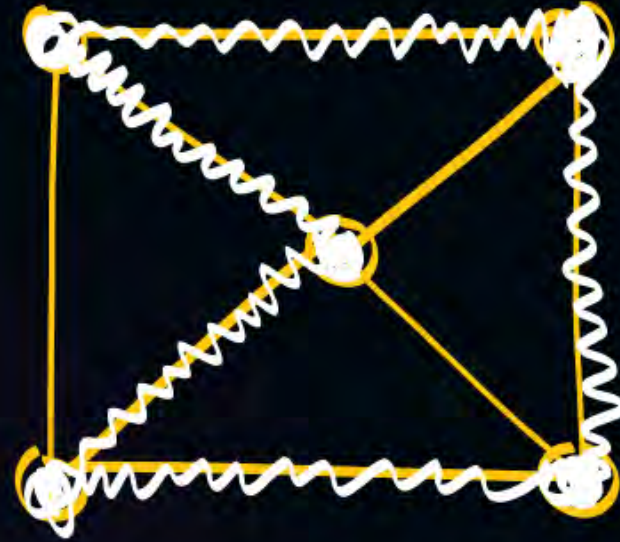
Hamiltonian Graph ✓

Euler Graph ✓

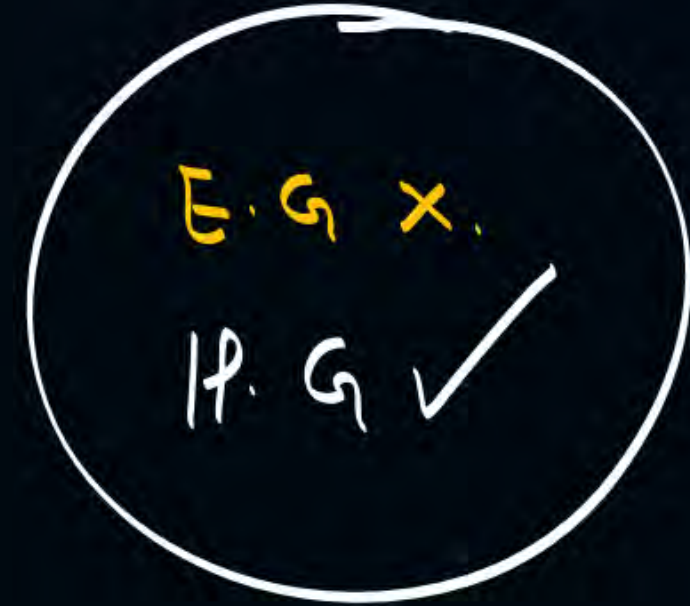
Wheel Graph:



H.G ✓



H.G ✓



K_1

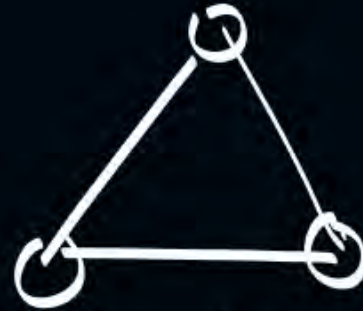
X



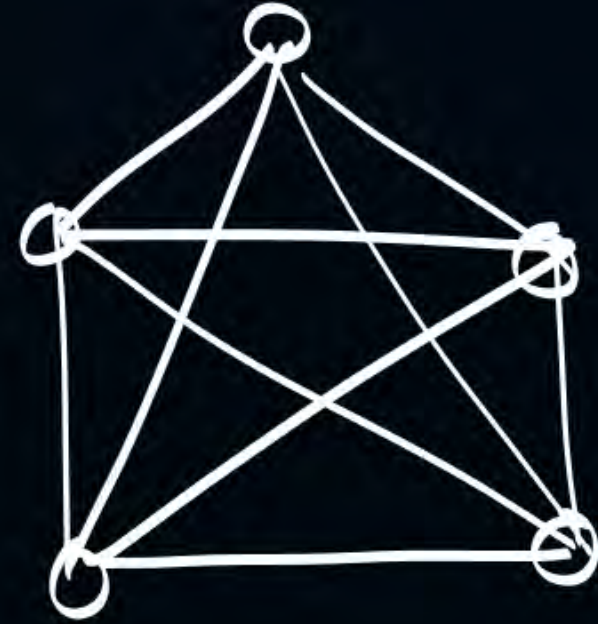
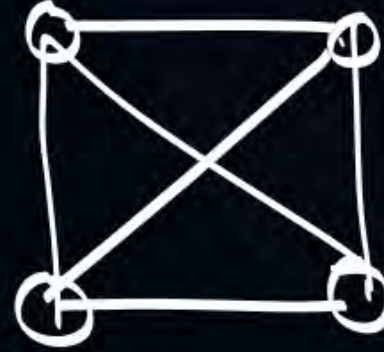
K_2

X

H.G X



K_3



K_n is Hamiltonian Graph ($n \geq 3$)

Complete bipartite. Graph:

H.G ✓

$K_{1,1}$



H.G ✗
H.P ✓

$K_{1,2}$



open.
H.G ✗
H.P ✓

$K_{2,3}$



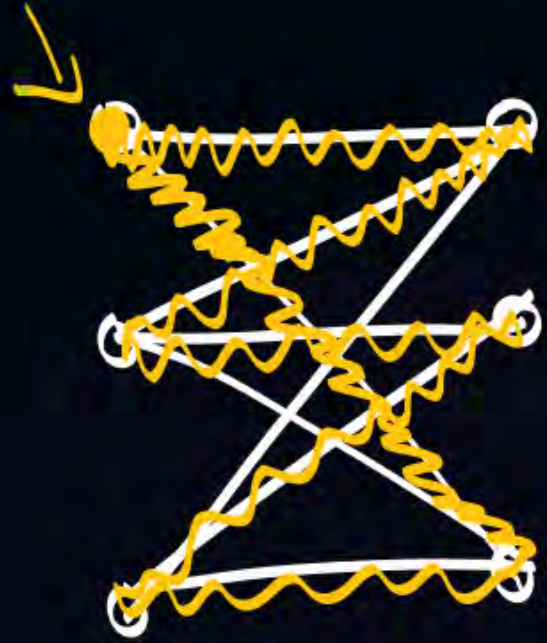
H.G ✗
H.P ✓

$K_{2,2}$



H.C ✓
H.P ✓

$K_{3,3}$



$K_{m,n}$ is Hamiltonian Graph ($m=n$)

$m, n \geq 2$

$K_{m,n}$ contain Hamiltonian path, $|m-n| \leq 1$.

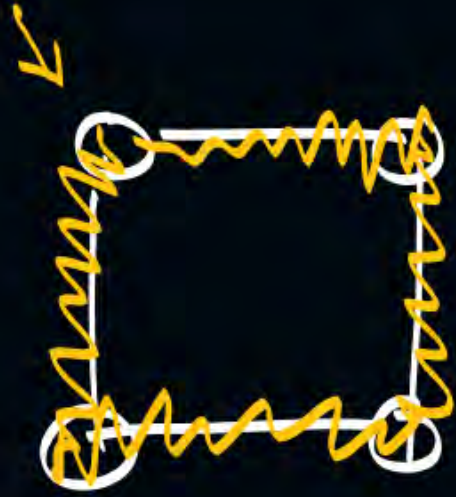
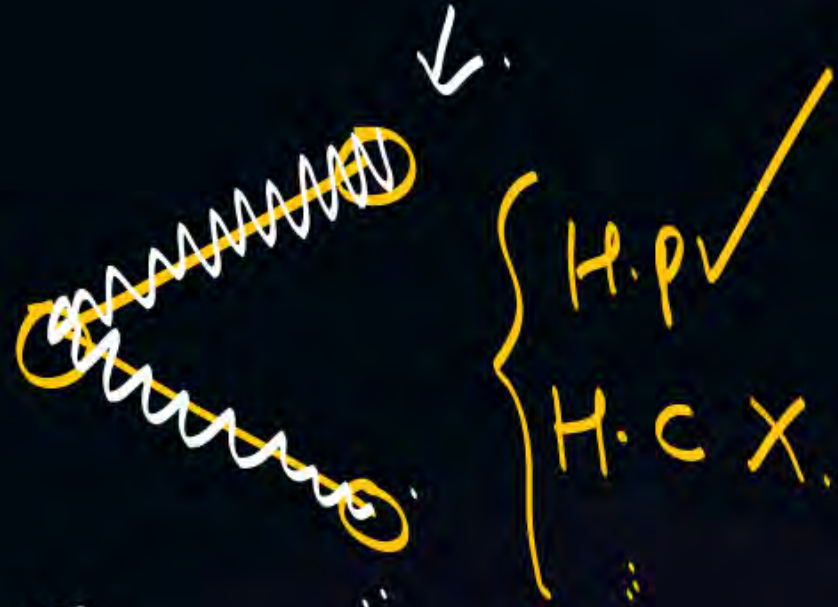
$K_{2,3}$

$|2-3| \leq 1$

$K_{2,2}$ $m=2$ $n=2$

H.C $\checkmark$ $|2-2| \leq 1$

H.P $\checkmark$ $\downarrow$ $\perp$



{ open
 +
 all vertices.

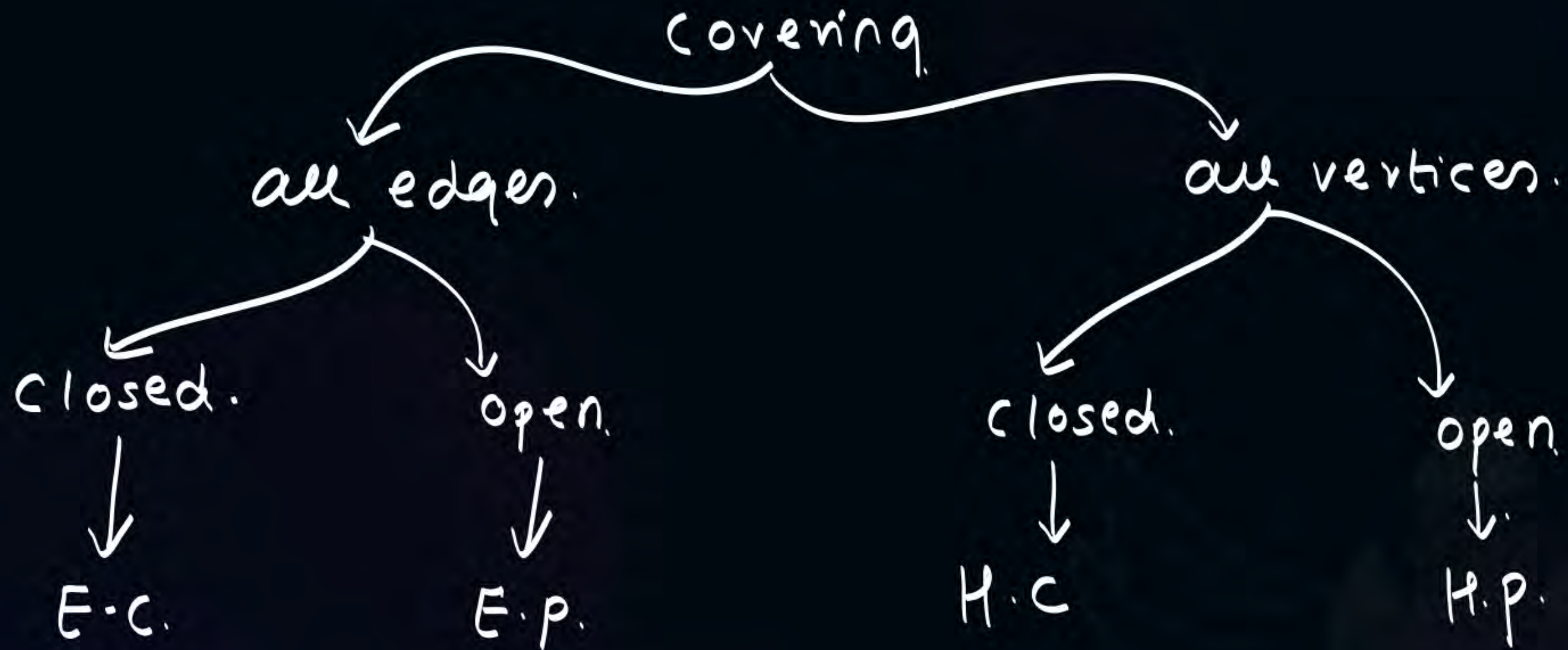
Hamiltonian path.

Every H.C contains H.P.

but H.P need not contains H.C.



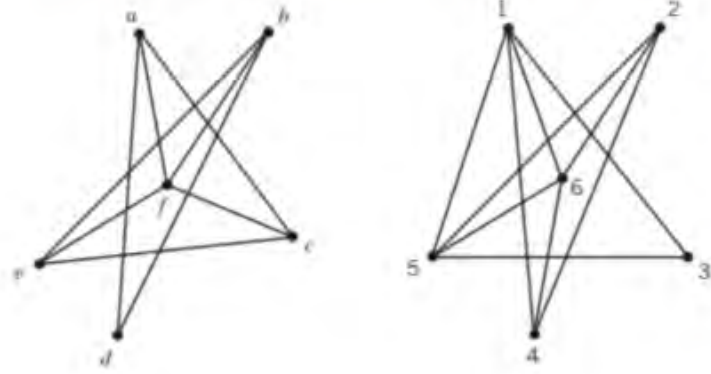
H.C. X.
 H.P. ✓



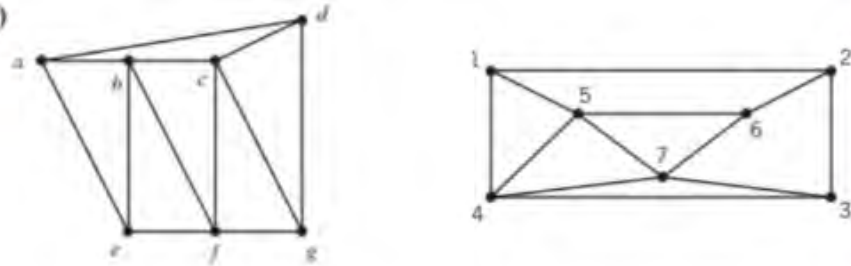
* Total no. of distinct Hamiltonian cycle in K_n .

* Total no. of distinct Hamiltonian cycle in $K_{n,n}$.
($n \geq 2$)

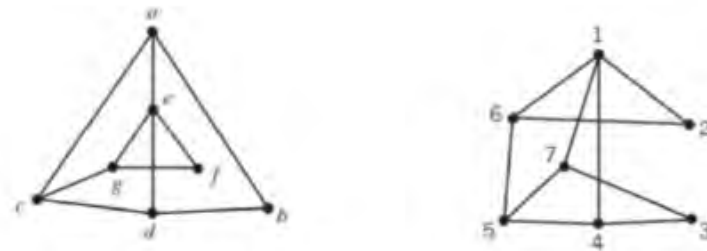
(e)



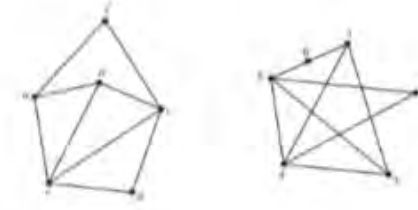
(f)



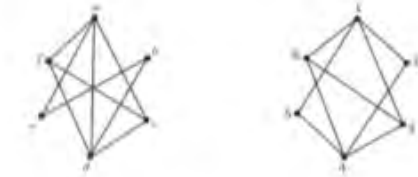
(g)



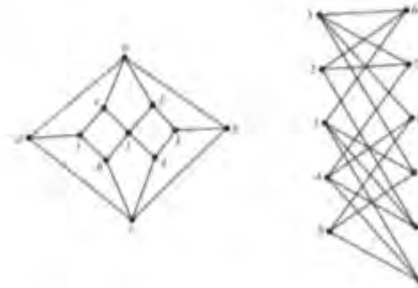
(i)



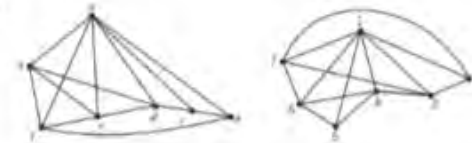
(j)



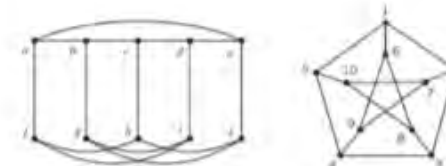
(k)



(h)



(l)





2 mins Summary

Topic

One

Connectivity in Graph Part-04

Topic

Two

Chromatic Number

Topic

Three

Topic

Four

Topic

Five



THANK - YOU